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#Create a simulator object
set ns [new Simulator]

#Open the nam trace file
set nf [open out.nam w]

$ns namtrace-all $nf

#Define a 'finish' procedure
proc finish { } {
    global ns nf
    $ns flush-trace
    #Close the trace file
    close $nf

    #Execute nam on the trace file
    exec nam out.nam &
    exit 0
}

#Create eight nodes
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
set n3 [$ns node]
set n4 [$ns node]
set n5 [$ns node]
set n6 [$ns node]
set n7 [$ns node]

#Create links between the nodes
$ns duplex-link $n0 $n3 1Mb 10ms RED
$ns duplex-link $n1 $n3 1Mb 10ms RED
$ns duplex-link $n2 $n3 1Mb 10ms RED
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$ns duplex-link $n3 $n4 1Mb 10ms RED
$ns duplex-link $n4 $n5 1Mb 10ms RED
$ns duplex-link $n4 $n6 1Mb 10ms RED
$ns duplex-link $n4 $n7 1Mb 10ms RED
$ns duplex-link-op $n0 $n3 orient right-up
$ns duplex-link-op $n3 $n4 orient middle
$ns duplex-link-op $n2 $n3 orient right-down
$ns duplex-link-op $n4 $n5 orient right-up
$ns duplex-link-op $n4 $n7 orient right-down
$ns duplex-link-op $n1 $n3 orient right
$ns duplex-link-op $n6 $n4 orient left

#Create a UDP agent and attach it to node n2
set udp0 [new Agent/UDP]
$ns attach-agent $n2 $udp0

#Create a CBR traffic source and attach it to udp0
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 500
$cbr0 set interval_ 0.005
$cbr0 attach-agent $udp0

#Create a Null agent (a traffic sink) and attach it to node n5
set null0 [new Agent/Null]
$ns attach-agent $n5 $null0

#Connect the traffic sources with the traffic sink
$ns connect $udp0 $null0

#Create a UDP agent and attach it to node n1
set udp1 [new Agent/UDP]
$ns attach-agent $n1 $udp1

#Create a CBR traffic source and attach it to udp1
set cbr1 [new Application/Traffic/CBR]
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$cbr1 set packetSize_ 500
$cbr1 set interval_ 0.005
$cbr1 attach-agent $udp1
#Create a Null agent (a traffic sink) and attach it to node n6
set null0 [new Agent/Null]
$ns attach-agent $n6 $null0
#Connect the traffic sources with the traffic sink
$ns connect $udp1 $null0
#Create a UDP agent and attach it to node n0
set udp2 [new Agent/UDP]
$ns attach-agent $n0 $udp2
#Create a CBR traffic source and attach it to udp2
set cbr2 [new Application/Traffic/CBR]
$cbr2 set packet size_ 500
$cbr2 set interval_ 0.005
$cbr2 attach-agent $udp2

#Create a Null agent (a traffic sink) and attach it to node n7
set null0 [new Agent/Null]
$ns attach-agent $n7 $null0
#Connect the traffic sources with the traffic sink
$ns connect $udp2 $null0
$udp0 set fid_ 1
$udp1 set fid_ 2
$udp2 set fid_ 3
#Define different colors for data flows
$ns color 1 Red
$ns color 2 Green
$ns color 2 Blue
```

#Schedule events for the CBR agents

\$ns at 0.1 "\$cbr0 start"

\$ns at 0.2 "\$cbr1 start"

\$ns at 0.5 "\$cbr2 start"

\$ns at 4.0 "\$cbr2 stop"

\$ns at 4.2 "\$cbr1 stop"

\$ns at 4.5 "\$cbr0 stop"

#Call the finish procedure after 5 seconds of simulation time

\$ns at 5.0 "finish"

#Run the simulation

\$ns run

OUTPUT:





